KW Supercapacitors

Coin cells



Photo is representative

Description

Eaton supercapacitors are unique, ultrahigh capacitance devices utilizing electric double layer capacitor (EDLC) construction combined with new, high performance materials. This combination of advanced technologies allows Eaton to offer a wide variety of capacitor solutions tailored to specific applications that range from a few microamps for several days to several milliamps for milliseconds.

All products feature low ESR for high power density with environmentally friendly materials for a green power solution. Eaton supercapacitors are maintenance-free with design lifetimes up to 20 years* and operating temperatures down to -40 °C and up to +85 °C.

Features and benefits

- High specific capacitance
- Low leakage current
- Long cycle life
- Eco-friendly
- Broad operating range ,full specification -40 °C to +85 °C

Applications

- Electric utility meters
- Motor control units
- Solar inverters
- Real-Time Clock (RTC) backup
- Programmable Logic Controllers (PLCs)
- Irrigation and water control systems

Environmental compliance

Pb HALOGEN HF FREE RoHS

* Supercapacitor lifetimes vary based on charge voltage and temperature. See Eaton's application guidelines or contact your local Eaton sales representative for more information on lifetime estimates

Powering Business Worldwide

Specifications[1]

Capacitance	0.1 F to 1.5 F
Maximum working voltage	5.5 V
Surge voltage[3]	6.3 V
Capacitance tolerance	-20% to +80% (+20 °C)
Operating temperature range[2]	-40 °C to +85 °C

Standard product

Capacitance (F)	Part number	Lead length	Maximum initial ESR (Ω) (Equivalent series resistance) measured @ 1 kHz	Typical mass (g)
0.1	KW-5R5C104-R	Standard	50	3.7
0.1	KW-5R5C104H-R	Short	50	3.7
0.22	KW-5R5C224-R	Standard	50	3.7
0.22	KW-5R5C224H-R	Short	50	3.7
0.33	KW-5R5C334-R	Standard	50	3.7
0.33	KW-5R5C334H-R	Short	50	3.7
0.68	KW-5R5C684-R	Standard	30	10.2
0.68	KW-5R5C684H-R	Short	30	10.2
1.0	KW-5R5C105-R	Standard	30	10.4
1.0	KW-5R5C105H-R	Short	30	10.4

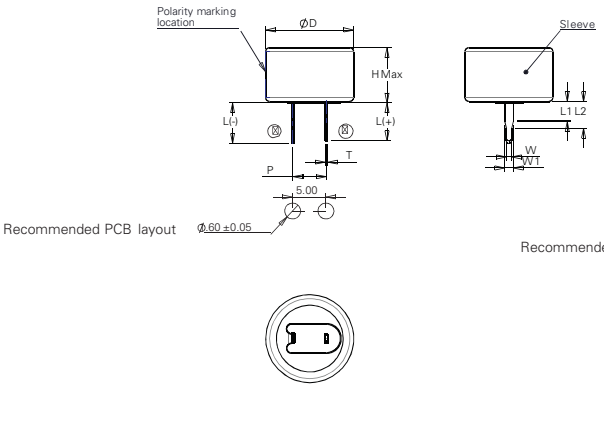
Performance

Parameter	Capacitance change (% of initial value)	ESR (% of maximum initial value)
Life — +85 °C @ 5.5 Vdc, 2000 hours	≤ 30%	≤ 200%
Storage Life — -40 °C to +85 °C, 2000 hours	≤ 30%	≤ 200%

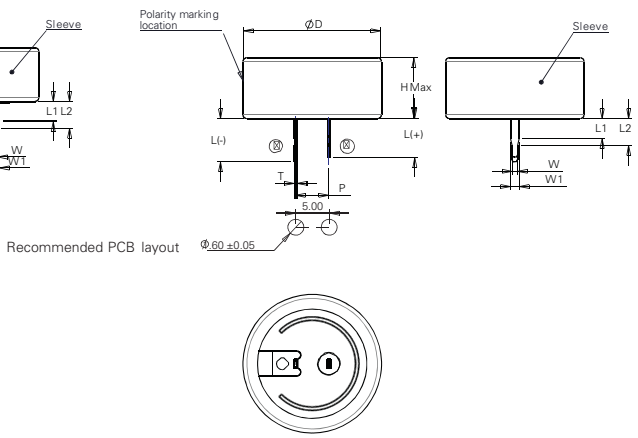
1. Testing and verification of product under end application conditions is recommended
2. Not recommended for +85 °C/85% RH applications
3. Surge voltage: Maximum voltage, non-repetitive, 1 second maximum

Dimensions (mm)

KW-5R5C104/224/334-R



KW-5R5C684/105-R



Part number	ØD maximum	H maximum	L (-) ±0.5	L (+) ±0.5	P ±0.3	T ±0.05	L1 ref	L2 ±0.5	W ±0.06	W1 ±0.06
KW-5R5C104-R	13.5	8.8	6.1	5.7	5.0	0.4	3.0	4.0	0.8	1.3
KW-5R5C104H-R	13.5	8.8	3.3	3.3	5.0	0.4	0.9	1.9	0.8	1.3
KW-5R5C224-R	13.5	8.8	6.1	5.7	5.0	0.4	3.0	4.0	0.8	1.3
KW-5R5C224H-R	13.5	8.8	3.3	3.3	5.0	0.4	0.9	1.9	0.8	1.3
KW-5R5C334-R	13.5	8.8	6.1	5.7	5.0	0.4	3.0	4.0	0.8	1.3
KW-5R5C334H-R	13.5	8.8	3.3	3.3	5.0	0.4	0.9	1.9	0.8	1.3
KW-5R5C684-R	21.5	9.5	6.7	6.0	5.0	0.4	3.2	4.2	0.8	1.3
KW-5R5C684H-R	21.5	9.5	3.5	3.5	5.0	0.4	1.0	2.0	0.8	1.3
KW-5R5C105-R	21.5	9.5	6.7	6.0	5.0	0.4	3.2	4.2	0.8	1.3
KW-5R5C105H-R	21.5	9.5	3.5	3.5	5.0	0.4	1.0	2.0	0.8	1.3

Part numbering system

KW	–5R5	C	47	4	H*	-R
Family code	Voltage (V) R = decimal	Configuration	Capacitance (µF) value	Multiplier	Short lead length	Standard product
KW = Family code	5R5 = 5.5 V	V = Vertical H = Horizontal C = Cylindrical	Example 474= 47 x 10^4 µF o r 0.47 F			

* If ordering standard lead length, omit "H" from part number

Packaging information

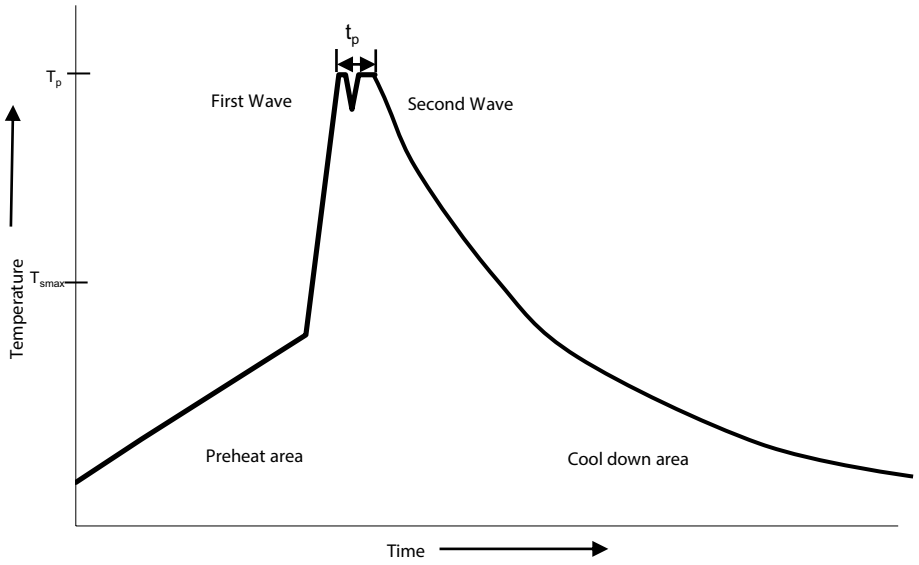
- KW-5R5C104/224/334-R—400 parts
- KW-5R5C684/105-R—100 parts

Part marking

- Manufacturer
- Capacitance (F)
- Maximum operating voltage (V)
- Polarity

Wave solder profile

WARNING: DO NOT EXCEED +100 °C BODYTEMPERATURE. PERMANENT DAMAGE MAY OCCUR



Profile Feature	Standard SnPb Solder	Lead (Pb) Free Solder
Preheat and soak • Temperature max. (T_{smax})	100 °C	100 °C
• Time max.	60 seconds	60 seconds
Δ preheat to max temperature	160 °C max.	160 °C max.
Peak temperature (Tp)*	235 °C – 260 °C	250 °C – 260 °C
Time at peak temperature (t_p)	5 seconds max	5 seconds max
Ramp-down rate	~ 2 K/s min ~3.5 K/s typ ~5 K/s max	~ 2 K/s min ~3.5 K/s typ ~5 K/s max
Time 25 °C to 25 °C	4 minutes	4 minutes

Manual solder

Do not touch the supercapacitor's external sleeve with the soldering rod or the sleeve will melt or crack. The recommended temperature of the soldering rod tip is less than +360 °C and the soldering duration should be less than 5 seconds. Minimize the time that the soldering iron is in direct contact with the terminals of the supercapacitor as excessive heating of the leads may lead to higher equivalent series resistance (ESR). Generally manual soldering is not recommended.

Reflow soldering

Do not use reflow soldering using infrared or convection oven heating methods.

Cleaning/Washing

Avoid cleaning of circuit boards, however if the circuit board must be cleaned use static or ultrasonic immersion in a standard circuit board cleaning fluid for no more than 5 minutes and a maximum temperature of +60 °C. Afterwards thoroughly rinse and dry the circuit boards. In general, treat supercapacitors in the same manner you would an aluminum electrolytic capacitor.

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